Appropriate models for data

1 For each of the following table of values:

i Plot the points from the table of values on a coordinate plane.

ii Determine whether the graph that best represents these values is linear, quadratic, or exponential.

iii Sketch the graph of the function represented by the table of values.

a

x	-1	0	1	2	3
y	7	4	1	-2	-5

b

x	-2	-1	0	1	2
y	18	6	2	6	18

c

x	-1	0	1	2
y	$2\frac{1}{3}$	3	5	11

d

x	-2	-1	0	1	2
y	$-\frac{7}{4}$	$-\frac{3}{2}$	-1	0	2

2 Sketch the graph of the function represented by the following tables of values:

a

x	-1	0	1	2	3
y	2	1	0	-1	-2

b

x	-2	-1	0	1	2
y	19	7	3	7	19

c

x	-1	0	1	2	3
y	$3\frac{1}{2}$	4	5	7	11

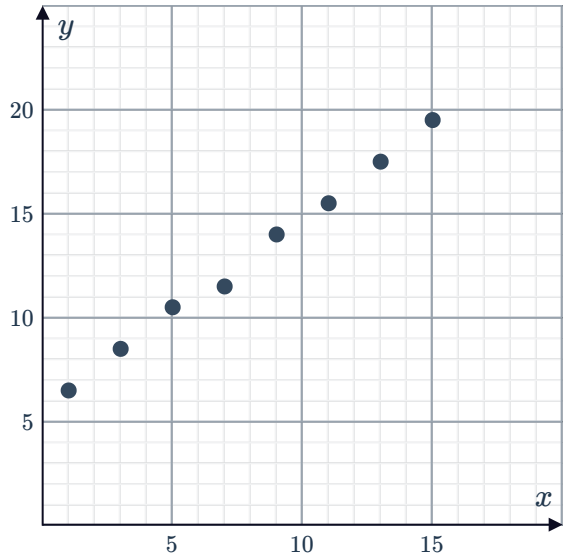
d

x	-2	-1	0	1	2
y	14	5	2	5	14

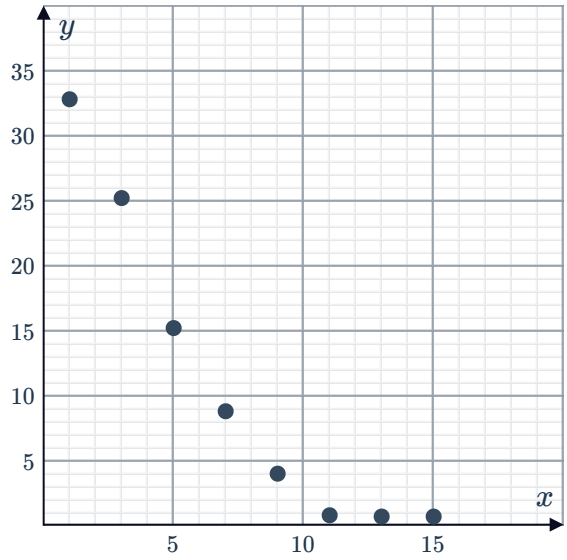
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3 For each of the following scatter plots, determine whether a linear model would be the best way to model the data and explain your answer:

a



b

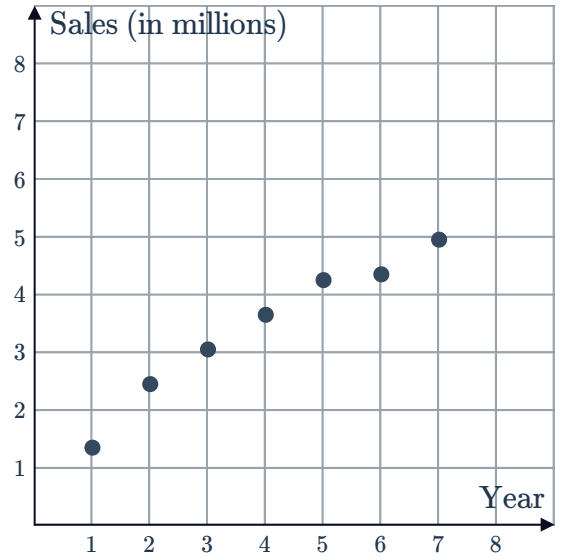


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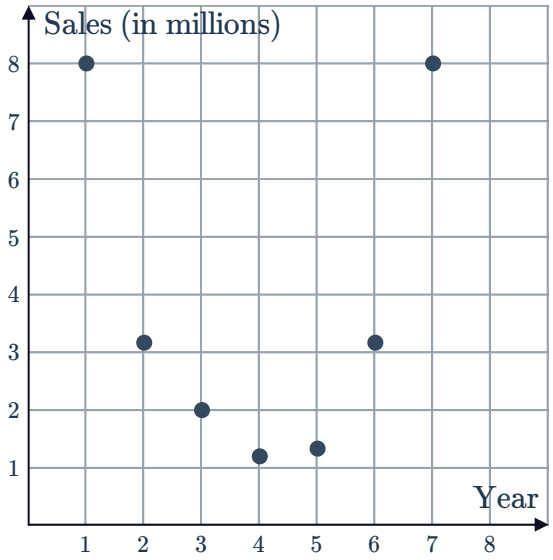
4 Select the most appropriate model for the data shown on each graph from one of the following:

- Quadratic, $f(x) = ax^2 + bx + c$, $a < 0$
- Quadratic, $f(x) = ax^2 + bx + c$, $a > 0$
- Linear, $f(x) = mx + b$
- Logarithmic, $f(x) = a \log x + c$

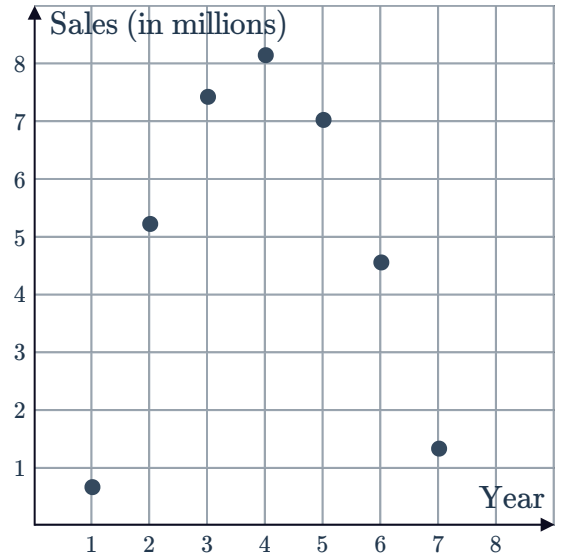
a



b



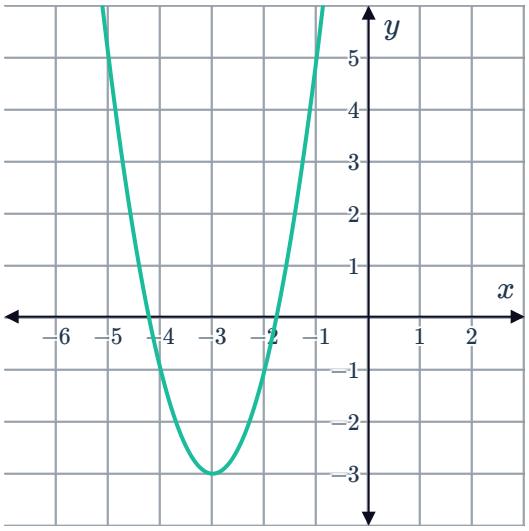
c



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Equations of models

5 Consider the following graph of a quadratic function:

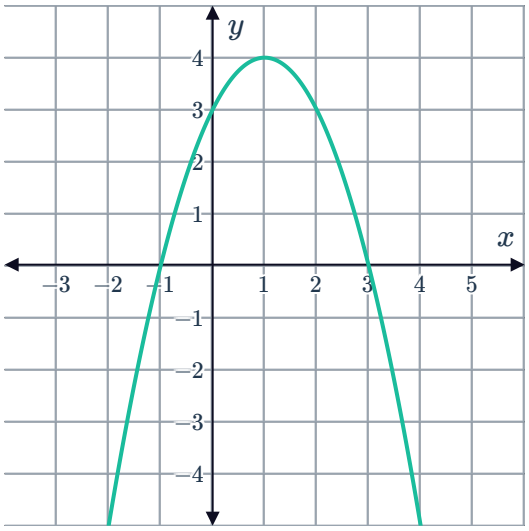


a Given that a quadratic function has a turning point at (h, k) and a is a constant, complete the standard equation form of the function:

$$y = a\,(x - \square)^2 + \square$$

b Determine the equation of the quadratic function.

6 Consider the following graph of a quadratic function:



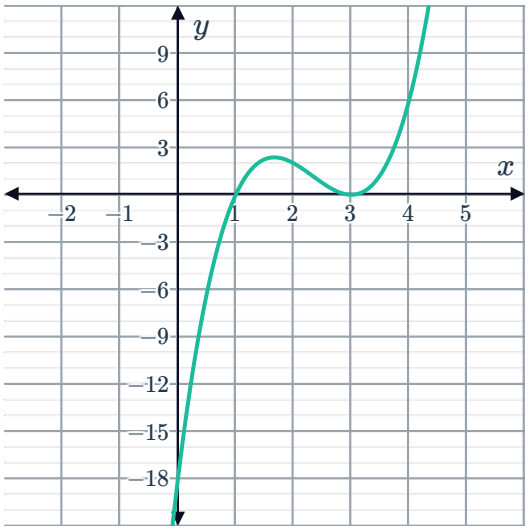
a Given that a quadratic function has roots at $x = p$ and $x = q$ and a is a constant, complete the standard equation form of the function:

$$y = a\,(x - \square)\,(x - q)$$

b Determine the equation of the quadratic function.

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7 Consider the following graph of a function:



a State the type of function this graph represents.

b Determine whether the equation of the function can be expressed in the following forms:

i $y = a(x-h)^2 + k$, where h, k and a are constants.

ii $y = a(x-p)(x-q)^2$, where p, q, r and a are constants.

iii $y = a(x-p)(x-q)$, where p, q and a are constants.

iv $y = a(x-p)(x-q)(x-r)$, where p, q, r and a are constants.

c Determine the equation of the function.

8 Consider the following graph of a function:



a State the type of function this graph represents.

b Determine whether the equation of the function can be expressed in the following forms:

i $y = a(x-h)^2 + k$, where h, k and a are constants.

ii $y = a(x-p)(x-q)(x-r)$, where p, q, r and a are constants.

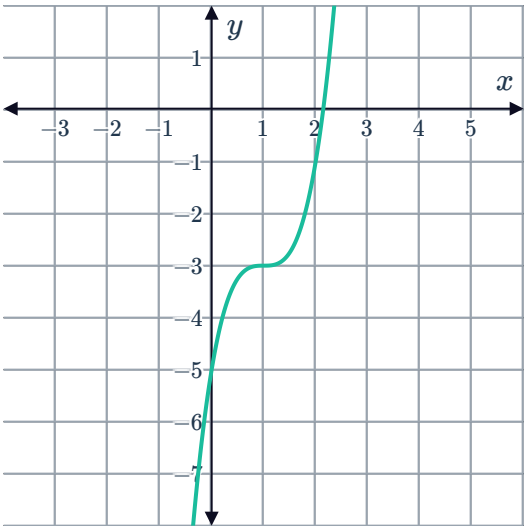
iii $y = a(x-p)(x-q)(x-r)(x-s)$, where p, q, r, s and a are constants.

iv $y = a(x-p)(x-q)$, where p, q and a are constants.

c Determine the equation of the function.

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9 Consider the following graph of a cubic function:

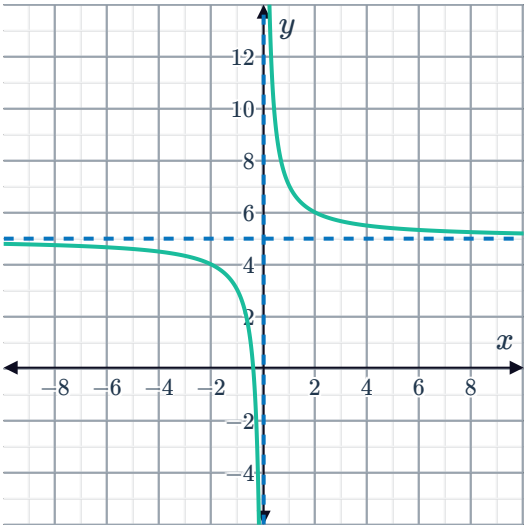


a Given that cubic function has a point of inflection at (h, k) and a is a constant, complete the standard equation form of the function:

$$y = a\,(x - \square)^3 + \square$$

b Determine the equation of the cubic function.

10 Consider the following graph of a function:



a State the type of function this graph represents.

b Determine whether the equation of the function can be expressed in the following forms:

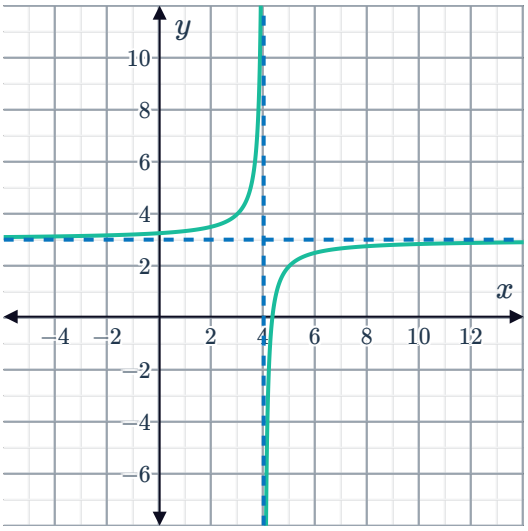
i $y = \dfrac{a}{x - h}$ ii $y = \dfrac{1}{x}$

iii $y = \dfrac{a}{x}$ iv $y = \dfrac{a}{x} + k$

c Determine the equation of the function.

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11 Consider the following graph of a function:



a State the type of function this graph represents.

b Determine whether the equation of the function can be expressed in the following forms:

i $y = \dfrac{a}{x-h} + k$

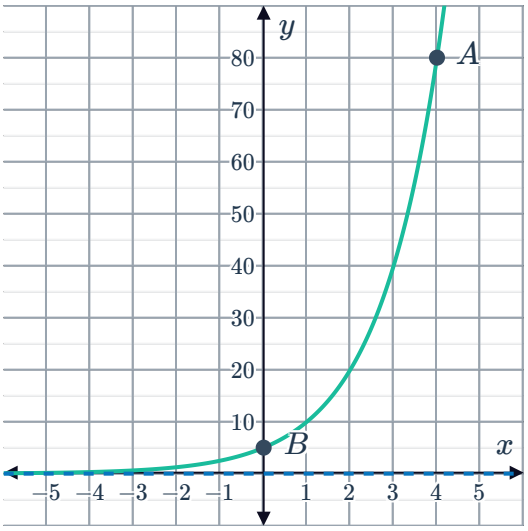
ii $y = \dfrac{1}{x}$

iii $y = \dfrac{a}{x-h}$

iv $y = \dfrac{a}{x}$

c Determine the equation of the function.

12 The points $A\,(4, 80)$ and $B\,(0, 5)$ lie on the exponential function that has been graphed:

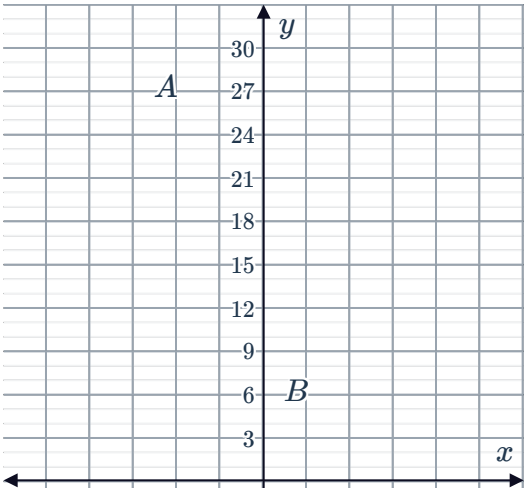


a Given the standard form of the exponential function $y = ab^x$, solve for the value of a.

b Now, find the value of b in the standard form $y = ab^x$.

c Now, state the equation of the function.

13 The points $A\,(-3, 27)$ and $B\,(0, 6)$ lie on the exponential function that has been graphed:



a Given the the standard form of the exponential function is $y = ab^{-x} + c$, find the value of:

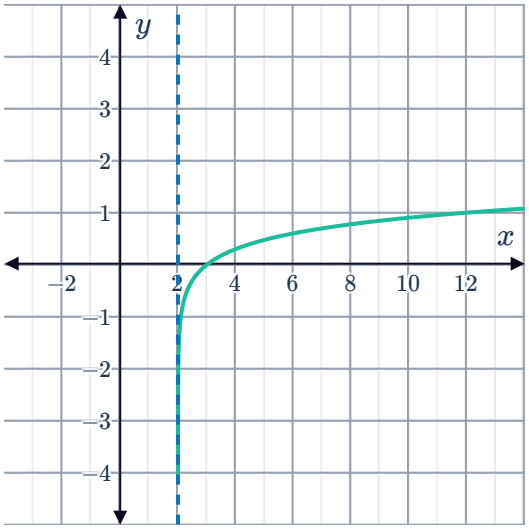
i c ii a iii b

b Now, state the equation of the function.

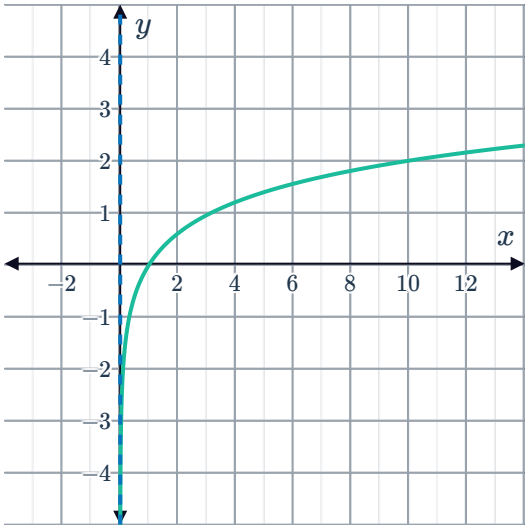
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14 For each of the following graphs, find the equation of the base 10 logarithmic function:

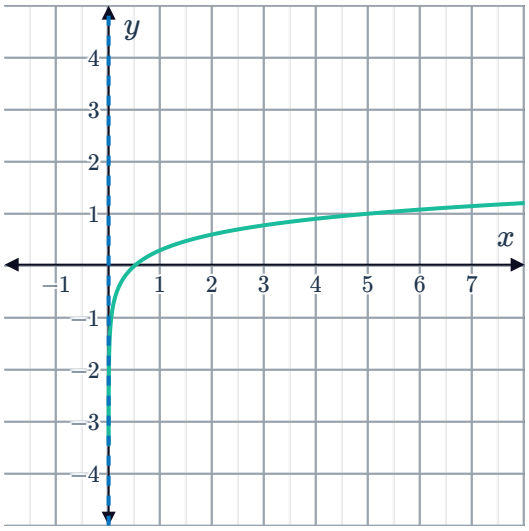
a



b



c



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15 For each of the following tables of values:

i State the type of function the table of values represent.

ii Find the equation of the function.

a

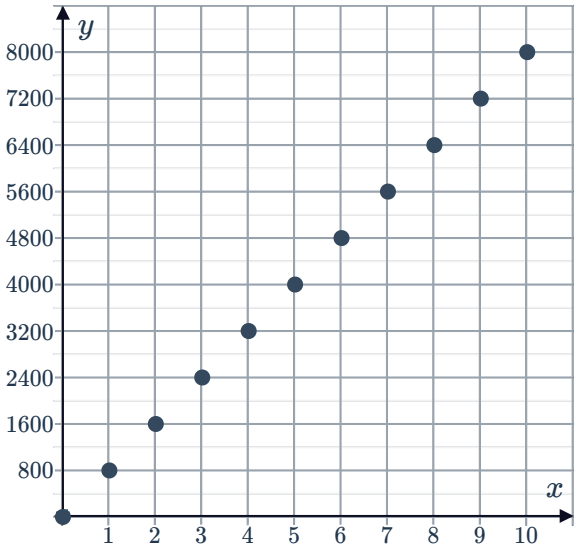
x	-12	-11	-10	-9	-8	-7	-6
y	7	$\frac{16}{3}$	$\frac{13}{3}$	4	$\frac{13}{3}$	$\frac{16}{3}$	7

b

x	1	2	3	4	16
y	16	8	$\frac{16}{3}$	4	1

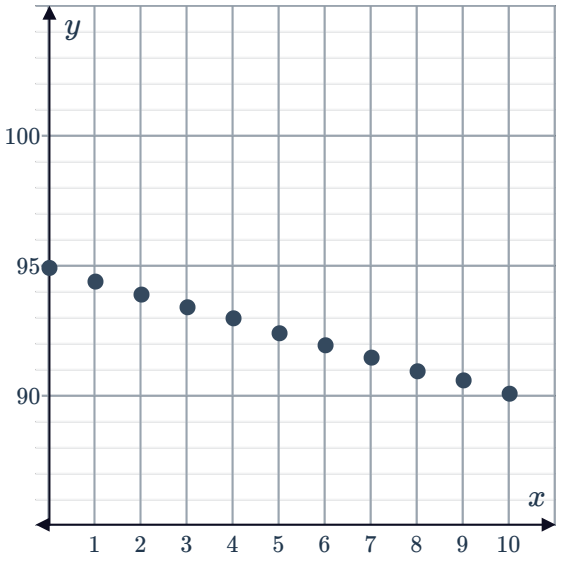
Applications

16 Every week, Rochelle deposits $\$800$ into a savings account. She plots her accumulated savings over several weeks as shown in the given graph.
Would a linear model accurately describe Rochelle's savings after the first week? Explain your answer.

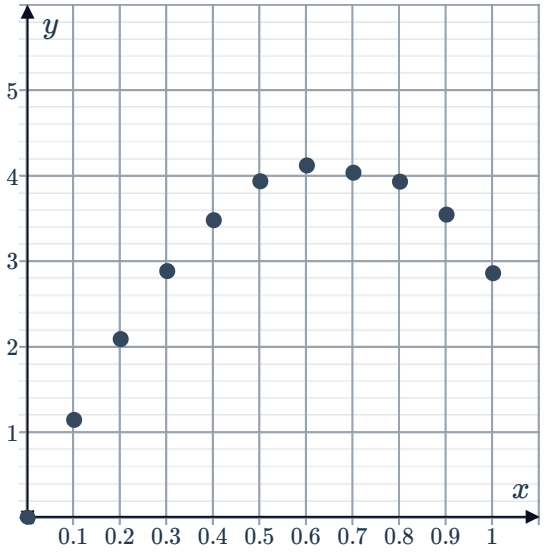


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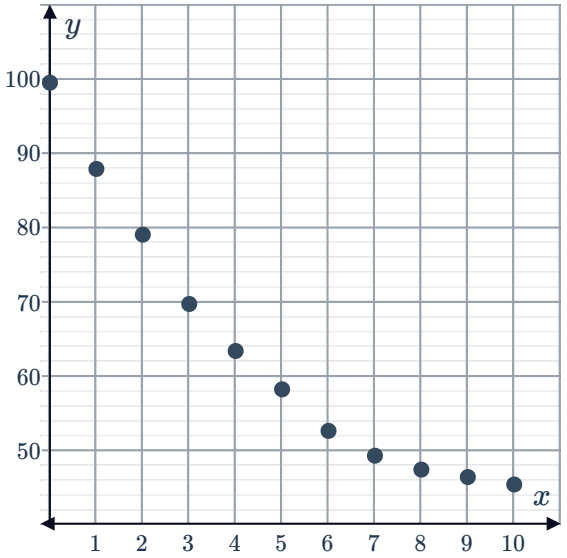
17 A large bucket has a very small leak near its base. The volume of water in the bucket, measured in cubic feet, is recorded once a minute and the results are plotted in the given graph.
Does a linear model accurately describe the volume of water over the total time plotted? Explain your answer.



18 A soccer ball is kicked at $20\ \mathrm{m/s}$ at a 50° angle. The height of the ball throughout the first second is plotted in the given graph.
Does a linear model accurately describe the height of the ball after being kicked? Explain your answer.



19 Mario cooks a pot of soup and measures the temperature as it cools down. The following graph shows the temperature of the soup, y , in degrees Celsius, measured at every minute, x.
Does a linear model accurately describe the temperature of the soup? Explain your answer.



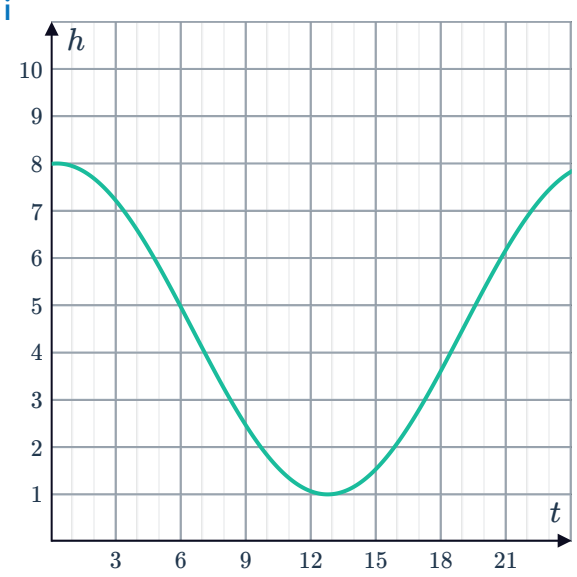
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20 A scientist wants to capture the rise and fall of the tide on a wharf pillar. On a particular day, the high tide occurred at midnight. He measures the height of the water every 6 hours for 24 hours.
The data is shown in the following table of values, where t is the time, in hours, after the initial observation, and h is the height of the water level, in meters, on the wharf pillar.

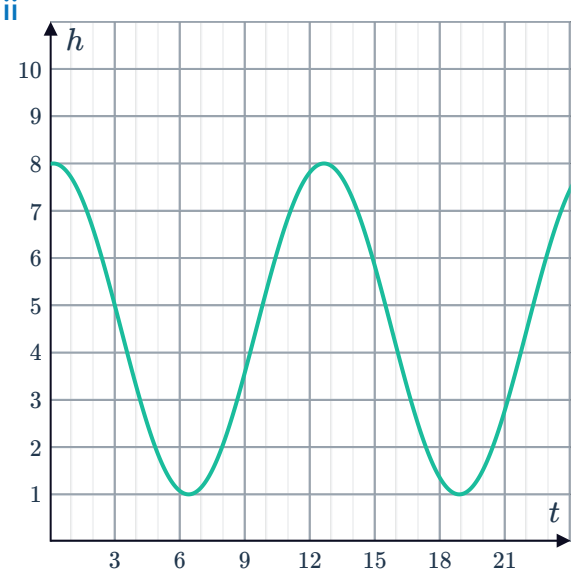
t	0	6	12	18	24
h	8.0	1.1	7.8	1.3	7.5

a Determine whether the following graphs match data:

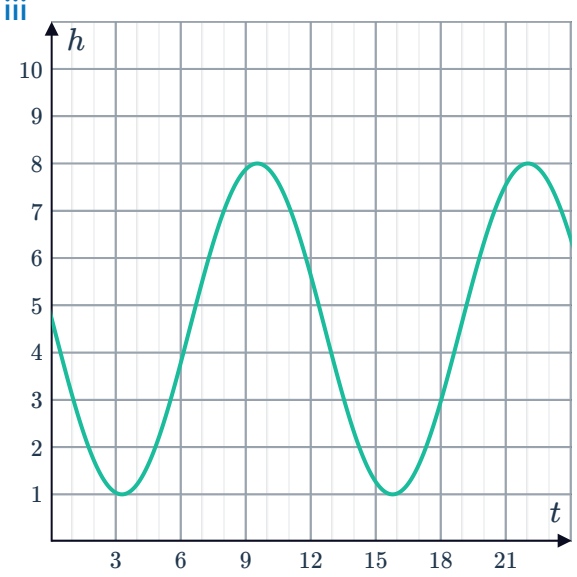
i



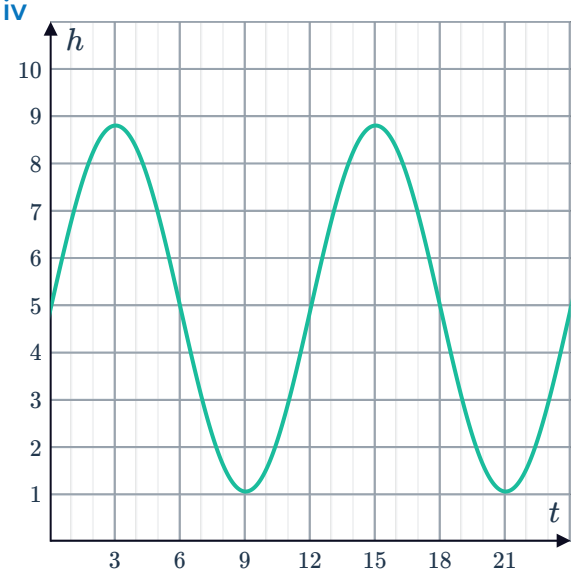
ii



iii



iv



b How many times over the 24-hour period was the high tide point reached?

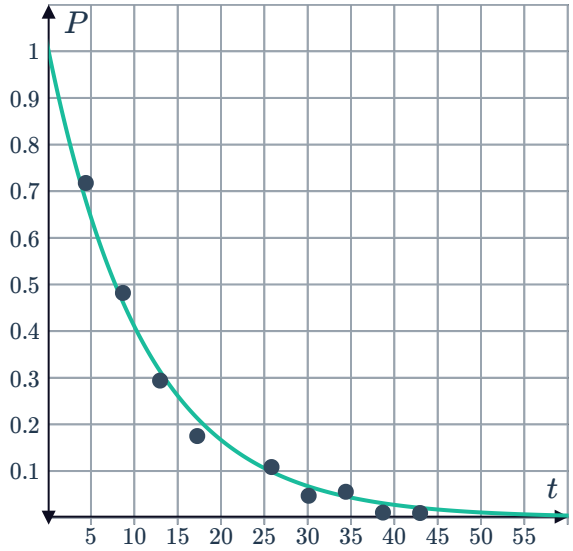
c What was the height of the water at low tide?

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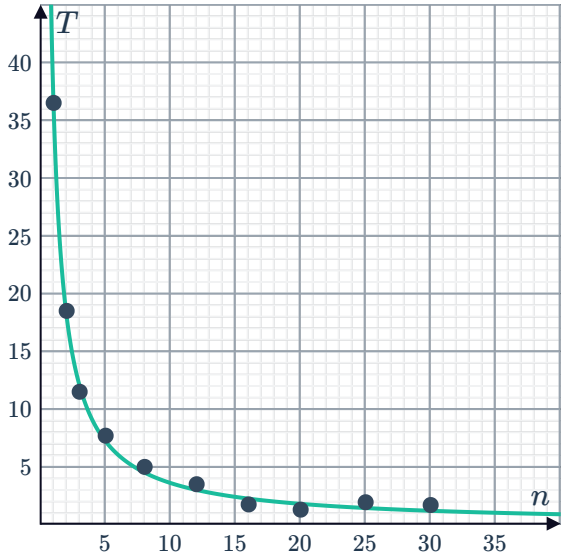
21 Due to a certain process of nuclear decay, the amount of Potassium-40 in a geological sample halves every 0.13 billion years. The scatter plot below compares the age, in billions of years, of a sample t with the proportion P of the original amount of Potassium-40 present in the sample, for several different geological samples. The graph also features an exponential model.

a As t gets larger and larger, what value does P approach?

b Consider a particular geological sample containing an initial amount of Potassium-40. As this sample ages, will the amount of Potassium-40 present always be greater than zero?



22 A building company is considering how many bricklayers to hire to build a wall. The company looks at their records and finds they have previously built 10 walls of a similar size, and have recorded how many hours, T, it took to build each wall and the number of bricklayers, n, used each time. To investigate the relationship between the two variables, an analyst plots the recorded data on a graph and models the time taken against the number of bricklayers using an inverse relationship.

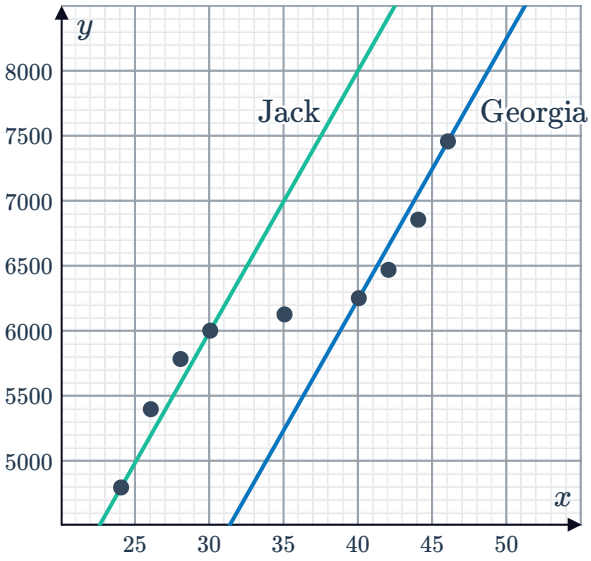


a As n gets larger and larger, what will happen to the value of T?

b Will the model still describe the build time if the company hires a very large number of bricklayers?

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23 A supermarket uses rectangular prism cardboard boxes of different widths, x, and volumes, y. A scatter plot in which the points represent different sized boxes is shown:



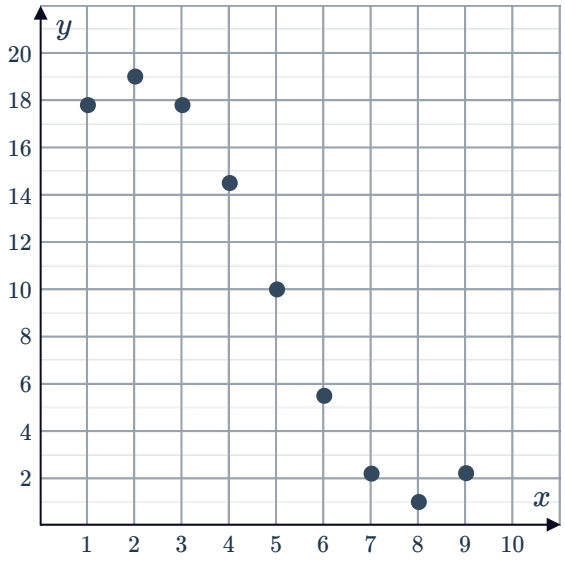
a Jack is an employee who usually packs and unpacks smaller boxes. He uses the linear model to the left on the scatter plot to represent the box sizes. State the interval for which this model is most accurate.

b Georgia is an employee who usually packs and unpacks larger boxes. She uses the linear model to the right on the scatter plot to represent the box sizes. State the interval for which this model is most accurate.

c Do the two models together accurately describe all of the boxes used by the supermarket? Explain your answer.

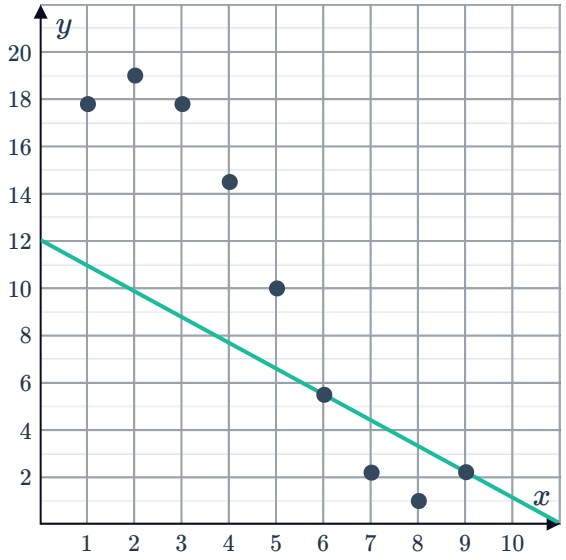
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24 A hydrologist arrives at a beach at 1 am and measures the level of the tide across 9 hours. The height of the tide, y, at every hour x is plotted in the given graph.

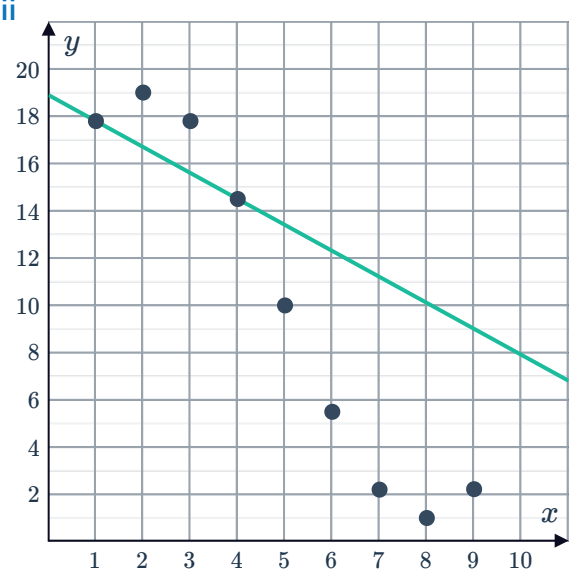


a Determine whether the following linear models would be most appropriate to model the ebbing tide, the period of time over over which the water level decreases:

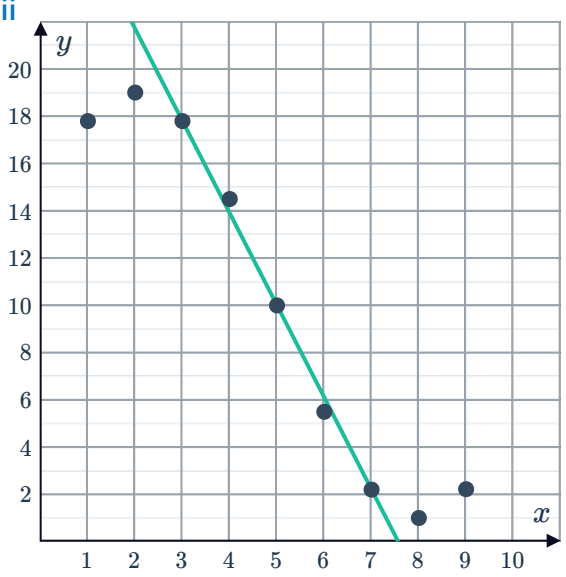
i



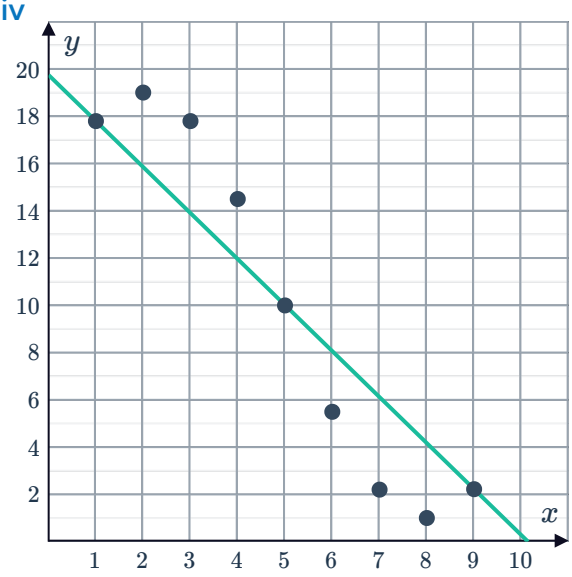
ii



iii



iv



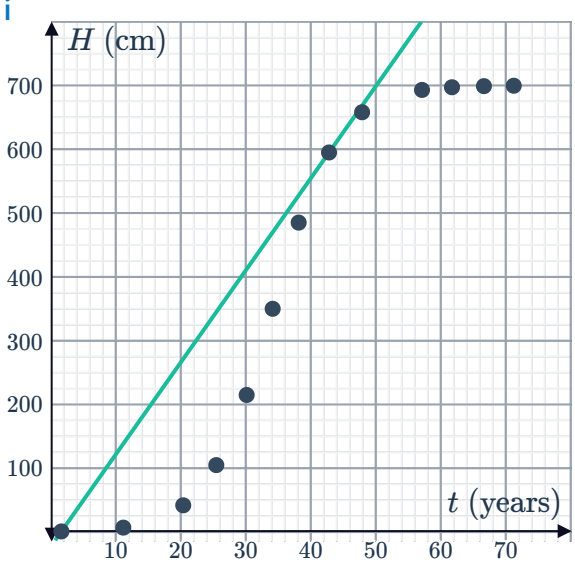
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b Should we be confident using the linear model to estimate the height of the tide at 2 pm? Explain your answer.

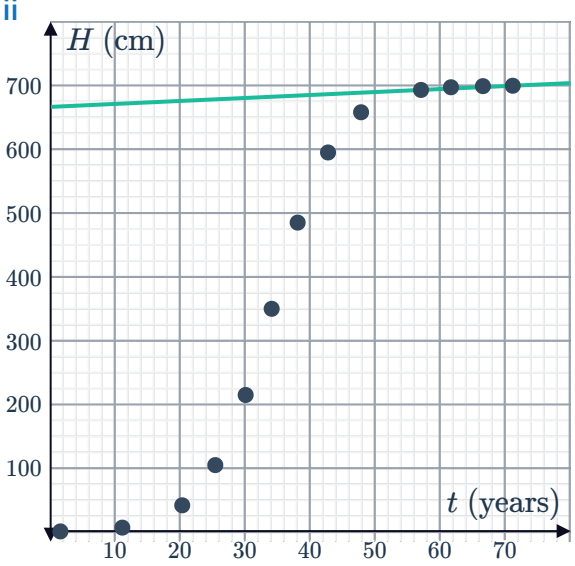
25 A forestry researcher is analysing the growth history of a single birch tree. He wants to use this data set to create a linear model that describes the height of any birch tree over the period of its most rapid growth.

a Determine whether the following linear models would be most appropriate to model the time period of most rapid growth:

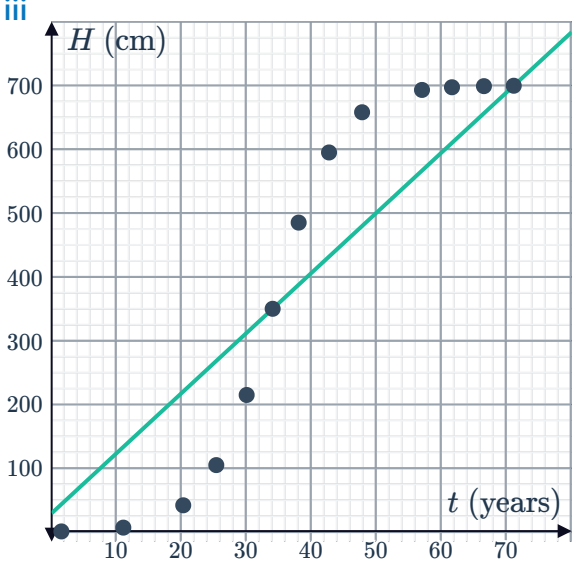
i



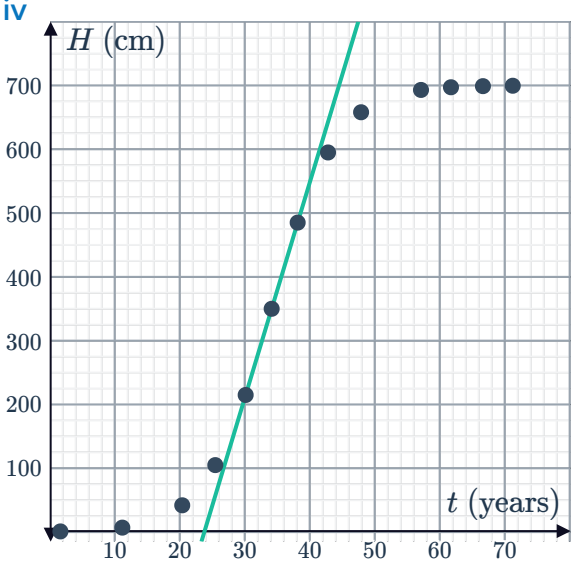
ii



iii



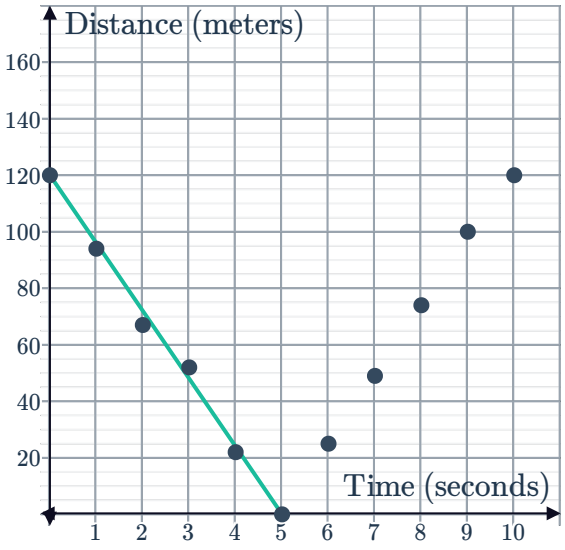
iv



b As the tree gets older, what will happen to its height?

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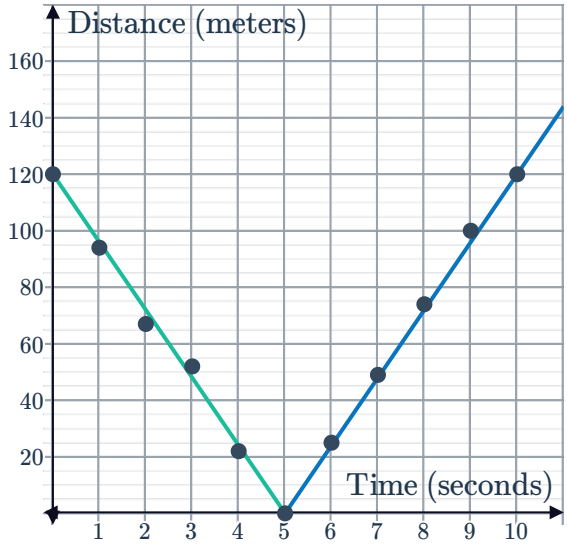
26 A truck travels along a highway which passes a town. The distance between the truck and the town over time is plotted in the given graph:



a At what point in time does the truck reach the town?

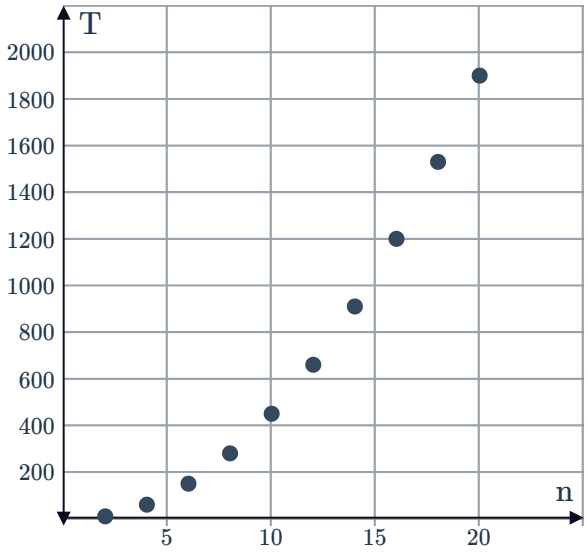
b Does the model fit the data before, after or before and after the truck reaches the town?

c The following piecewise linear model is used to better model the distance:
Is all the data accurately described by the piecewise linear model?



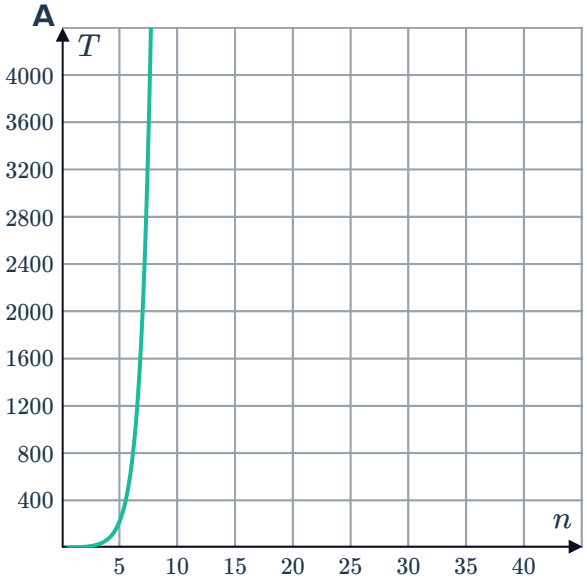
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27 A computer program compares and orders the scores of all students who sit an exam. The time taken, T in milliseconds, for the program to completely order all students is plotted for different numbers of students, n:

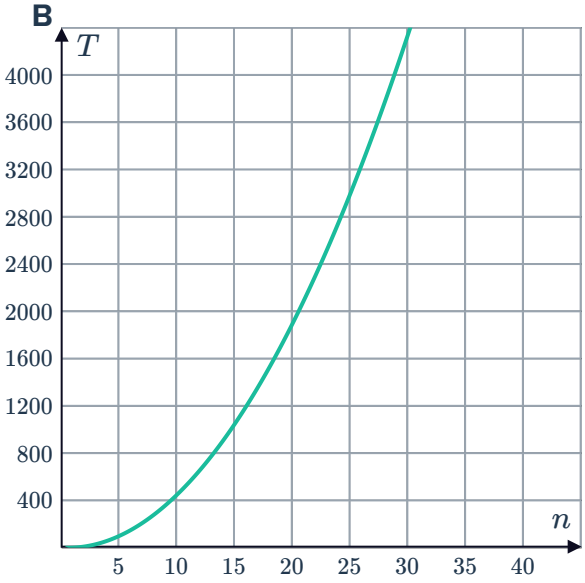


Which one of the following model will best describe the run time of the computer program as the number of students continues to increase?

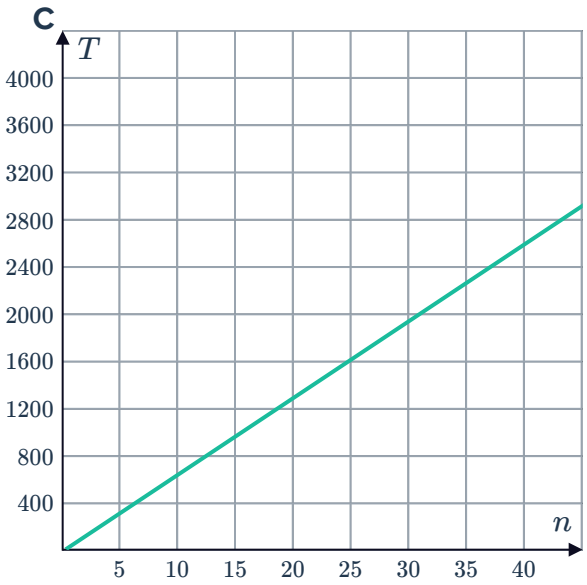
A



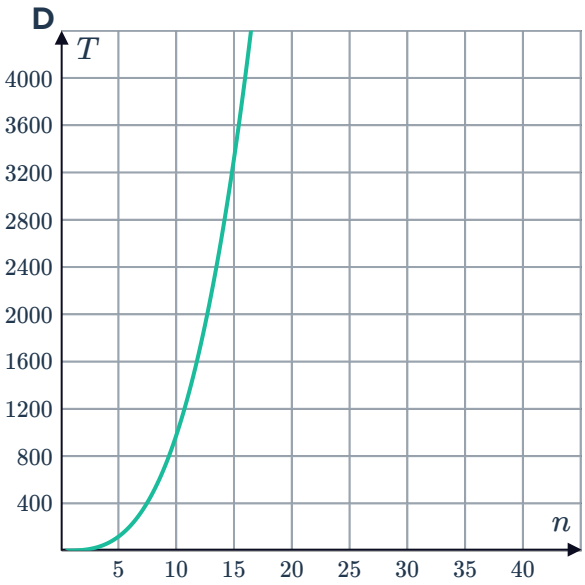
B



C

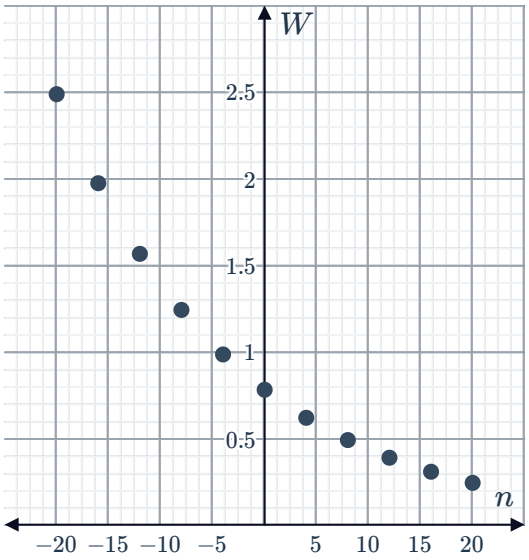


D



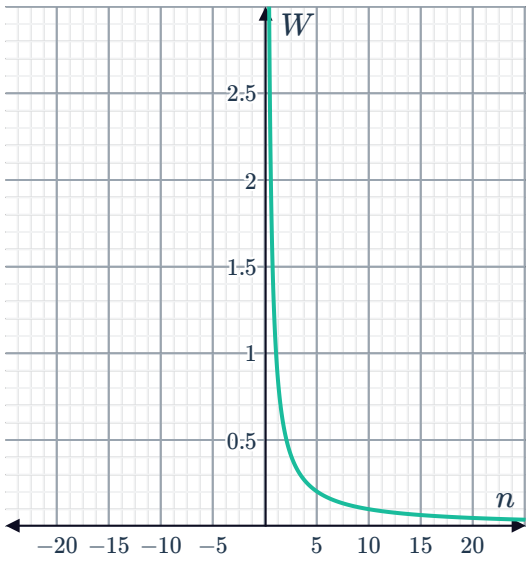
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28 The musical note A440 is often used to tune instruments. It has a frequency of $440\ \text{Hz}$ and a wavelength of approximately $0.784\ \text{m}$. Hermione wants to know the wavelength, W, of some notes, n, above and below A440. She measures the wavelength of these notes played on her violin and graphs the results in the scatter plot:

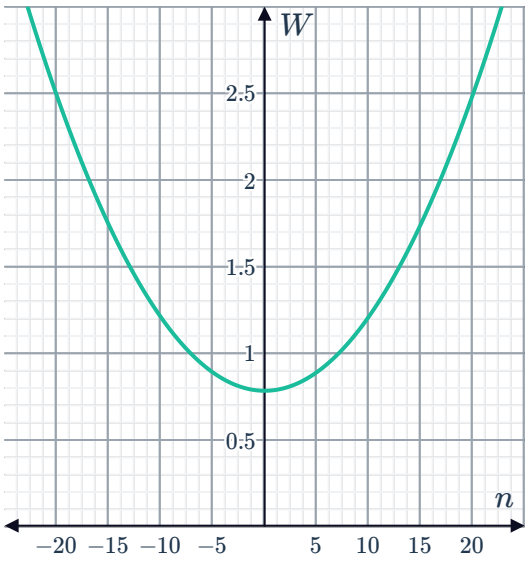


Which model will best describe the wavelength of notes far away from A440?

A



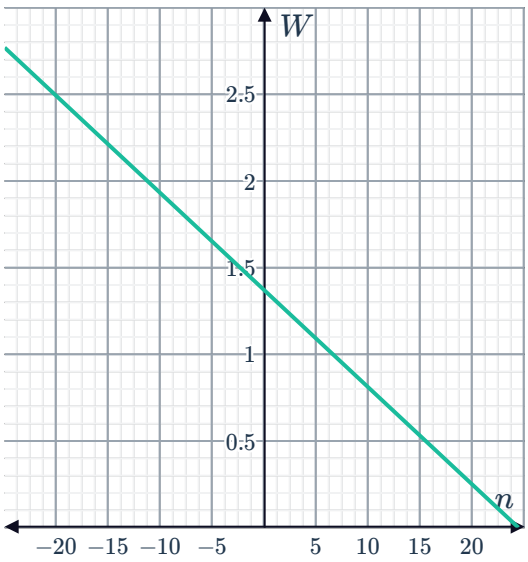
B



C



D



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29 The growth of algae, A, in a pond over time t can be modeled with the following equation:

$$A = 1.4e^{0.49t}$$

a Complete the table of values, correct to one decimal place.

b Sketch the graph of the function.

A	0	1	2	3
t				

30 The following table shows the approximate total number of patent applications in the U.S. for a number of years since 1900:

a Create a scatter plot using the data in the table.

b Determine if the data suggests an exponential association. Explain your answer.

Year	Applications (in thousands)
2010	480
2005	370
1995	260
1980	170
1950	80
1900	40

31 The following table shows the increase in temperature over time in a closed car on a hot day:

a Create a scatter plot using the data in the table.

b State whether a linear, exponential, or logarithmic function would best model the data.

Time (minutes)	Temperature (°C)
0	29
10	40
20	45
30	48
40	50
50	52
60	53

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